

Lecture 2: Evaluation Metrics

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The Scenario: You built a Logistic Regression to detect **Credit Card Fraud**.

- Dataset: 1,000,000 transactions.
- Fraud rate: 0.1% (1,000 frauds, 999,000 legit).

The Result: Your model achieves **99.9% Accuracy**.

Is this a good model?

The "Dumb" Model: Imagine a model that simply predicts "**Legit**" (0) for absolutely everyone.

- Correct predictions: 999,000 (all the legit ones).
- Wrong predictions: 1,000 (all the frauds).
- **Accuracy:** $\frac{999,000}{1,000,000} = \mathbf{99.9\%}$.

Conclusion: In the context of **Imbalanced Datasets** (Fraud, Cancer, Spam), Accuracy is a misleading metric. It rewards the model for doing nothing. *We need deeper metrics to understand "How" the model is wrong.*

The Confusion Matrix: Dissecting Performance

Instead of a single score, we look at the 4 possible outcomes.

		Predicted Class	
		Positive (1)	Negative (0)
Actual Class	Positive (1)	True Positive (TP)	False Negative (FN)
	Negative (0)	False Positive (FP)	True Negative (TN)

- **TP:** We predicted Fraud, and it was Fraud. (Success)
- **TN:** We predicted Legit, and it was Legit. (Success)
- **FP:** We predicted Fraud, but it was Legit. (False Alarm)
- **FN:** We predicted Legit, but it was Fraud. (Missed Detection)

In hypothesis testing (and Data Science interviews), errors have specific names:

Type I Error (False Positive)

- "False Alarm".
- Rejection of a true null hypothesis.
- *Example:* Convicting an innocent person.

Type II Error (False Negative)

- "Missed Detection".
- Failing to reject a false null hypothesis.
- *Example:* Letting a guilty person go free.

Memorization Tip: The boy who cried wolf caused a Type I error (False Alarm) first, then the village committed a Type II error (Missed the wolf).

Errors are not created equal. In business, **FN** $\neq$ **FP**.

Example: Cancer Detection

- **Cost of FP (Type I):** Patient stress, cost of extra biopsy. (Moderate)
- **Cost of FN (Type II):** Patient dies. (Catastrophic)

Example: YouTube Copyright Filter

- **Cost of FP:** Creator gets angry, appeals. (Customer Service cost)
- **Cost of FN:** YouTube gets sued by Universal Music. (Legal cost)

*As a Data Scientist, you must tune the model to minimize the **Total Business Cost**, not just maximize Accuracy.*

Question: "When the model predicts YES, how often is it right?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Focus:** Quality of the positive predictions.
- **High Precision means:** Low False Positive rate. You rarely cry wolf.
- **Use Case: Email Spam Filter.**
 - User hates checking the Spam folder.
 - If you mark an email as Spam (Prediction=1), you better be sure.
 - FP (Deleting a work email) is unacceptable.

Question: "Out of all the real YESs, how many did we find?"

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Focus:** Quantity / Coverage.
- **High Recall means:** Low False Negative rate. You rarely miss a target.
- **Use Case: Terrorist Detection / Ebola Screening.**
 - You cannot afford to let a single case pass through.
 - FP (Stopping an innocent person) is annoying but acceptable.
 - FN (Missed terrorist) is catastrophic.

Often overlooked, but crucial in medicine. **Question:** "Out of all the Healthy people (Negatives), how many did we correctly identify as healthy?"

$$\text{Specificity} = \frac{TN}{TN + FP}$$

The Trade-off:

- High Recall usually implies Lower Specificity.
- If you arrest everyone (Recall = 100%), your Specificity is 0% (you arrested all innocents too).

It is usually impossible to maximize both simultaneously.

Scenario: Strict Model

- "Only predict 1 if probability $> 90\%$ "
- Result: **High Precision, Low Recall.**
- (We miss many, but the ones we find are sure).

Scenario: Lax Model

- "Predict 1 if probability $> 10\%$ "
- Result: **Low Precision, High Recall.**
- (We find everyone, but generate many false alarms).

Are they the same? **No.** They answer two completely different questions.

Precision (Vertical View)

- Focus: **Predictions**.
- Denominator: All Predicted Positives ($TP + FP$).
- *"Of all the people I arrested, how many were guilty?"*

Specificity (Horizontal View)

- Focus: **Reality (Negatives)**.
- Denominator: All Actual Negatives ($TN + FP$).
- *"Of all the innocent people, how many did I let go?"*

The Golden Rule:

- **Recall** (aka Sensitivity) is accuracy on the **Actual Positives**.
- **Specificity** is accuracy on the **Actual Negatives**.
- **Precision** is the quality of the **Positive Prediction**.

Let's visualize the difference with a simple analogy. **The Scenario:** A nightclub has a line of 100 people:

- **50 VIPs** (Positives).
- **50 Non-VIPs** (Negatives).

The "Lax" Bouncer: He lets absolutely **everyone** in.

1. Precision (Purity inside)

- Question: "Of the people inside, are they mostly VIPs?"
- Calculation: $\frac{50 \text{ VIPs}}{100 \text{ People inside}} = \mathbf{50\%}$
- *Result: Mediocre (Coin flip).*

2. Specificity (Filtering at the door)

- Question: "Of the Non-VIPs, did you stop them?"
- Calculation: $\frac{0 \text{ Stopped}}{50 \text{ Non-VIPs}} = \mathbf{0\%}$
- *Result: Catastrophic (No filter).*

Conclusion: Precision measures the *outcome quality*, Specificity measures the *process quality* (the filter).

We need a single metric to find the balance. Why not the arithmetic mean $\frac{P+R}{2}$?

- Example: Precision=1.0, Recall=0.0 (Useless model). Arithmetic Mean = 0.5.
(Misleading)

The Harmonic Mean (F1-Score):

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Properties: If either P or R is low, the F1-score crashes close to 0.
- It forces the model to be good at **both**.

Definition of Harmonic Mean: The inverse of the *average of the inverses*.

$$H = \frac{1}{\text{Arithmetic Mean}(\frac{1}{P}, \frac{1}{R})}$$

Step-by-Step Derivation:

1. Average the inverses:

$$\frac{\frac{1}{P} + \frac{1}{R}}{2} = \frac{\frac{P+R}{P \cdot R}}{2} = \frac{P+R}{2 \cdot P \cdot R}$$

2. Take the inverse of the result to get F1:

$$F_1 = \frac{1}{\frac{P+R}{2 \cdot P \cdot R}} = \frac{2 \cdot P \cdot R}{P+R}$$

Deep Dive: Why do we invert the numbers?

The Harmonic Mean is mandated by math whenever you average **Ratios** with a **Fixed Numerator**.

1. The Analogy: Speed

$$\text{Speed} = \frac{\text{Distance (Fixed)}}{\text{Time (Variable)}}$$

To average speed over a fixed distance, you must average the *Times* (the denominators). Since $\text{Time} = \frac{D}{\text{Speed}}$, you are averaging the inverses.

2. The F1-Score Context

$$\text{Precision} = \frac{\text{TP}}{\text{Predicted Volume}} \quad \text{Recall} = \frac{\text{TP}}{\text{Actual Volume}}$$

- Both metrics share the same goal: capturing **TP**.
- We want to average the "Volume required to get 1 TP" (the inverse).
- **Result:** Averages of inverses → Harmonic Mean.

What if Business says: "Recall is 2x more important than Precision"? We use the generalized F_β score:

$$F_\beta = (1 + \beta^2) \times \frac{\text{Precision} \times \text{Recall}}{(\beta^2 \times \text{Precision}) + \text{Recall}}$$

Strategic Choice of β :

- $\beta = 1$: Standard F1 (Balance).
- $\beta = 0.5$: Focus on **Precision** (e.g., Spam, Recommender Systems).
- $\beta = 2$: Focus on **Recall** (e.g., Cancer, Fraud).

The Confusion: If $\beta = 2$ means "Recall is important", why does the formula have $\beta^2 \times \text{Precision}$?

$$F_{\beta} = (1 + \beta^2) \frac{\text{Precision} \times \text{Recall}}{(\beta^2 \times \text{Precision}) + \text{Recall}}$$

The Explanation (Cross-Multiplication): F_{β} is a weighted Harmonic Mean. We apply the weight β^2 to the inverse of Recall ($\frac{1}{R}$).

$$\text{Weighted Sum} = \frac{1}{P} + \beta^2 \cdot \frac{1}{R}$$

To add these fractions, we must cross-multiply:

$$\frac{1 \cdot R + \beta^2 \cdot P}{P \cdot R}$$

Algebraically, the weight of Recall ends up multiplying Precision because of the common denominator.

Logistic Regression outputs a probability $P \in [0, 1]$. We need a binary decision $\hat{y} \in \{0, 1\}$.

$$\hat{y} = 1 \iff P > \text{Threshold}$$

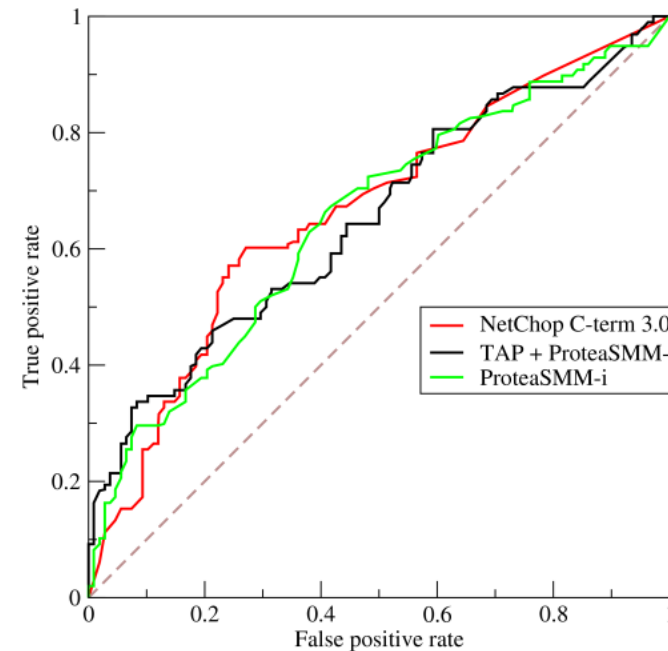
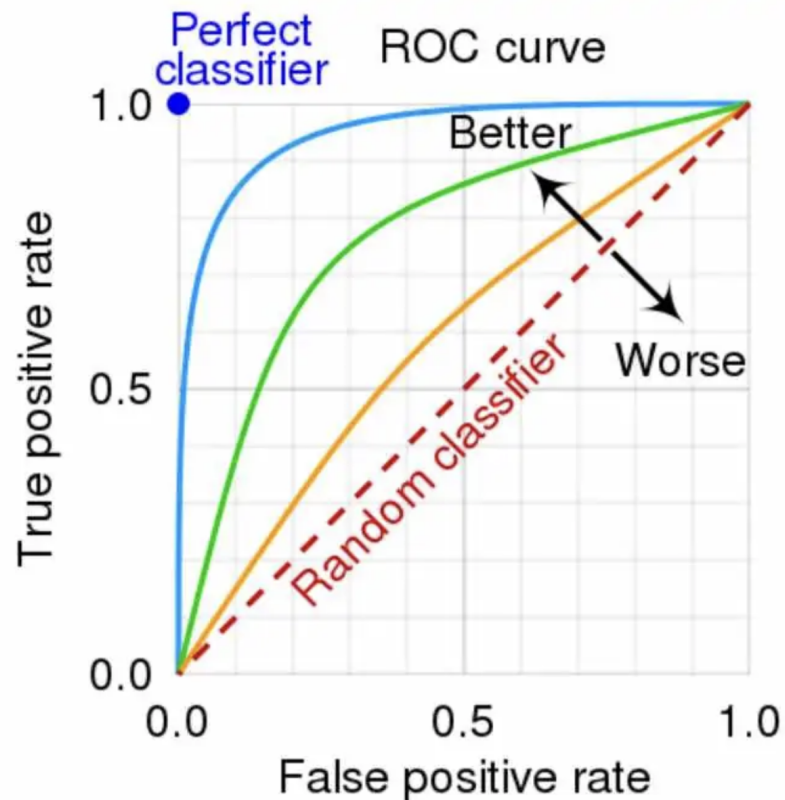
- **Default Threshold:** 0.5.
- **Moving the Threshold:**
 - Increase Threshold (e.g., 0.8) $\rightarrow$ Harder to predict 1 $\rightarrow$ Precision $\uparrow$, Recall $\downarrow$.
 - Decrease Threshold (e.g., 0.2) $\rightarrow$ Easier to predict 1 $\rightarrow$ Precision $\downarrow$, Recall $\uparrow$.

We need a way to evaluate the model regardless of the threshold choice.

ROC Curve (Receiver Operating Characteristic)

We plot the performance for **ALL possible thresholds**.

- **X-Axis:** False Positive Rate ($1 - \text{Specificity}$). "Risk of False Alarm".
- **Y-Axis:** True Positive Rate (Recall). "Ability to Detect".



- **Top-Left Corner:** Ideal ($\text{TPR}=1, \text{FPR}=0$).
- **Diagonal Line:** Random Guessing.

How do we compare two ROC curves? We calculate the Area Under the Curve.

- **AUC = 0.5:** Random model (no discrimination capacity).
- **AUC = 1.0:** Perfect model.
- **Typical Good Model:** 0.8 - 0.9.

Probabilistic Interpretation (Deep Dive): AUC is the probability that the model ranks a randomly chosen Positive instance higher than a randomly chosen Negative instance.

$$AUC = P(\text{score}(x^+) > \text{score}(x^-))$$

It measures the ranking quality, not the calibration.

The Blind Spot of AUC: Ranking vs Calibration

AUC only cares about **Order**. It does not care about the **Value**.

Comparison: Two models predicting the same Fraud ($Y = 1$) vs Legit ($Y = 0$).

Model	Score(Fraud)	Score(Legit)	Check ($>$)	AUC Contribution
Model A (Confident)	0.95	0.05	Yes	+1 Point
Model B (Compressed)	0.51	0.49	Yes	+1 Point

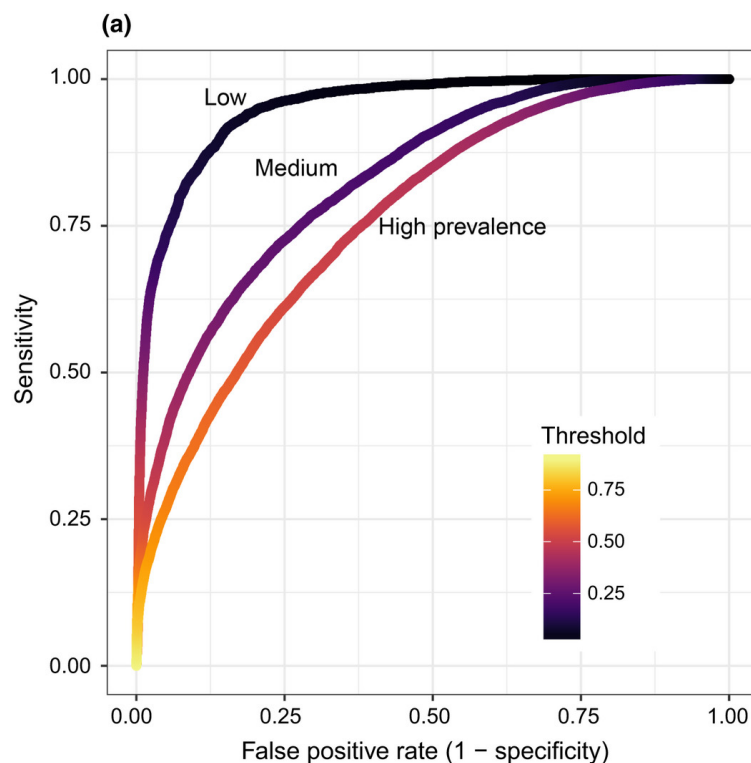
The Consequence:

- Both models have the **same AUC** (Perfect Ranking).
- **Model A** is well calibrated: 0.95 means "High Danger".
- **Model B** is poorly calibrated: 0.51 looks like a "Coin Flip", even though it correctly identified the fraud.

*AUC tells you **who** is riskiest. Calibration tells you **how risky** they are.*

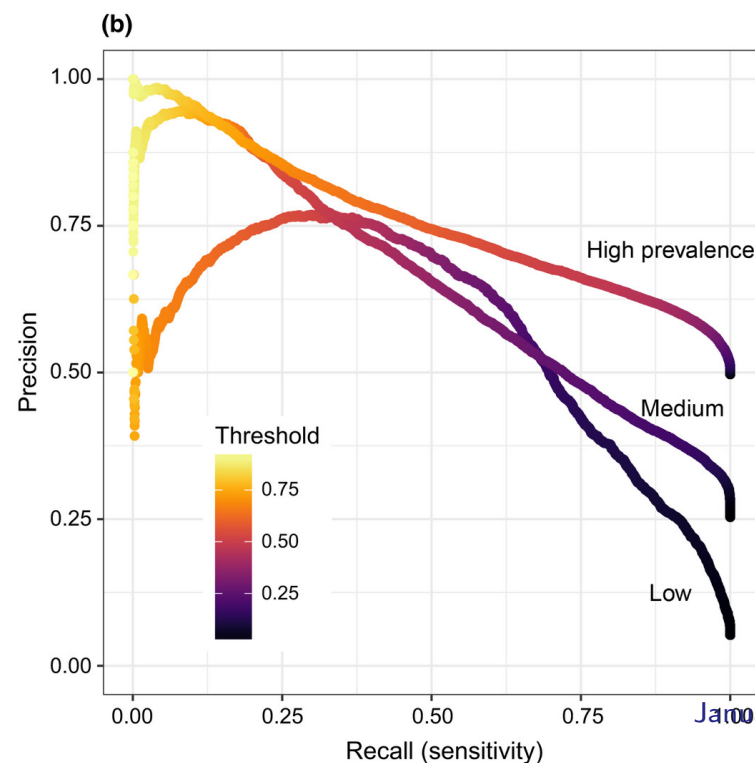
ROC Curve

- Uses TPR and FPR.
- **Robust to imbalance:** Does not change if we add 1M negatives.
- Use when negatives are roughly equal to positives, or when TN matters.



Precision-Recall Curve

- Uses Precision and Recall.
- **Sensitive to imbalance:** If class 1 is rare, Precision drops fast.
- **Gold Standard** for highly imbalanced data (Fraud, Anomaly detection).



The ROC Curve tells the story of a trade-off: "How much noise must I accept to find the signal?"

The Axes:

- **Y-Axis (Benefit):** True Positive Rate (Recall). We want this to go up fast.
- **X-Axis (Cost):** False Positive Rate (Risk of False Alarm). We want this to stay low.

The Ideal Shape (Top-Left):

- Ideally, the curve shoots straight up (finding positives) before moving right (making mistakes).
- **AUC = 1.0:** The model creates a perfect separation. It finds all positives before making a single mistake.

The Baseline (Diagonal): Unlike PR, the "Random" baseline is always the diagonal ($y = x$, $AUC=0.5$). If your curve touches this line, your model provides zero information.

Unlike ROC (where we aim Top-Left), here we aim for the **Top-Right**.

The Narrative: As we move Right (increasing Recall/Coverage), does the Precision (Quality) stay high, or does it crash?

Key Landmarks:

- **Top-Right Corner (1,1):** The "Holy Grail". Perfect Recall with Perfect Precision.
- **The "Cliff":** The point where Precision starts to drop significantly. We want to push this cliff as far right as possible.
- **The Baseline (Random):** A horizontal line at $y = \frac{\text{Positives}}{\text{Total}}$. (e.g., 0.01 for Fraud).

Comparison Metric: AUC-PR (Average Precision).

- 1.0 = Perfect.
- Baseline = % of Positives (e.g. 0.01). *Beating this is the real test.*

How do we calculate Precision/Recall if we have 3 classes (A, B, C)? We compute the metric for each class individually (One-vs-Rest), then average them. **Two ways to**

average:

1. Macro Average

- Calculate F1 for A, F1 for B, F1 for C.
- Take the simple average.
- **Effect:** Treats all classes equally. Useful if you care about the minority class performance.

2. Weighted Average

- Average weighted by the number of true instances in each class.
- **Effect:** Dominated by the majority class. Closer to "Global Accuracy".

Trap: In imbalanced multi-class, 'Weighted' can hide poor performance on rare classes. Use 'Macro' to be strict.

Dataset: Credit Card Fraud (Highly Imbalanced).

1. **Baseline:** Train a Logistic Regression. Calculate Accuracy. (Expected: $\approx 99\%$).
2. **Real Metrics:** Calculate the Confusion Matrix, Precision, Recall, and F1.
3. **Threshold Tuning:**
 - Predict probabilities (`.predict_proba`).
 - Create a loop to calculate F1-score for thresholds from 0.1 to 0.9.
 - Find the threshold that maximizes the F1-score.
4. **Visualization:** Plot the ROC Curve and calculate AUC.
5. **Business Decision:**
 - Assume $\text{Cost}(\text{FN}) = \1000 (Stolen money) and $\text{Cost}(\text{FP}) = \10 (Client call).
 - Find the threshold that minimizes Total Cost.